

Visualization Techniques for Precise Alignment in VR: A Comparative Study

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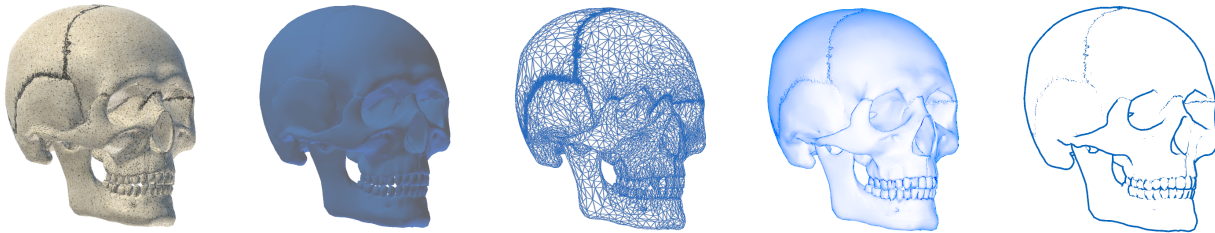


Figure 1: Screenshots of the four visualization techniques employed to compare users' performance for precise object alignment in 6 DoF. From left to right: Dynamic textured object and its four representations rendered with the visualization techniques –*Semitransparent*, *Wireframe*, *Fresnel-Derivative*, and *Silhouette*–.

ABSTRACT

Many studies explored the effectiveness of augmented, virtual, and mixed reality for object placement tasks. Two main approaches for assisting users during object alignment exist: static visualization techniques and interactive guides. This paper presents a comparative evaluation of four static visualization techniques used to render virtual objects when precise alignment in 6 degrees of freedom (DoF) is required. The selection of these techniques is based on the amount of occlusion caused by the visual guides during the alignment task. To the best of our knowledge, no previous work exists that evaluates which visualization technique is most suitable to support users while precisely aligning objects in virtual environments. We designed a virtual reality scenario considering two conditions –with and without time constraints– in which users aligned pairs of objects. To evaluate the users performance, quantitative and qualitative scores were collected. Our results suggest that visualization techniques with low levels of occlusion can improve alignment performance and increase user acceptance.

Index Terms: Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies; Human-centered computing—Visualization—Visualization design and evaluation methods—; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Precise object alignment is an activity performed frequently in processes related to assembly, manufacturing, maintenance, assistance, and training. In these scenarios, users have to place objects of interest in a specific position and orientation as a requirement to complete the tasks. Many studies have tried to prove the potential of using Augmented, Virtual, and Mixed Reality (AR, VR, and MR respectively) in environments that require to perform these kind of activities. Works by Caudell and Mizell [5], Feiner et al. [7], Reiners et al. [19], Henderson and Feiner [8], demonstrated the usability of their systems by recreating the scenarios in which users had to per-

form the alignment tasks, while some others such as those presented by Tang et al. [21], Robertson et al. [20], and Khuong et al. [12], use assembly blocks as objects of interest.

A common process in the aforementioned applications relies in the fact that visual cues are generated and displayed to assist users during task performance. Two main approaches are commonly used to represent these cues: static visualization techniques in which virtual replicas of real objects are typically rendered using transparent or wireframe representations; and interactive systems that use guides such as arrows, animations, or text, to indicate users the action to be performed.

In this work, we focus on analyzing static visualization techniques since they can be used in activities where precise tracking of the objects that need to be aligned is not available, but their corresponding Computer-Aided Design (CAD) models exist.

A common way to assist users during alignment tasks using static visualization techniques, is by rendering virtual models as solid objects to indicate the position in which a real object has to be placed [12, 19, 21]. This kind of representation leads to a high level of occlusions when the real and virtual objects overlap. Similar techniques modify the transparency of virtual models to allow users to see through them when overlap occurs [4, 16, 20, 23, 24]. Alternative approaches reduce the amount of occlusion rendering the virtual objects as wireframes by connecting a subset of the model vertices using lines [5, 7, 12]. While this technique presents some advantages in comparison to the solid or semitransparent representations, like reducing significantly the amount of occlusion when the objects overlap, many details of the real objects remain occluded to the users' view.

Some other approaches represent the virtual objects as line drawings showing the edges of the models and additional features that provide visual cues to the user [6, 11]. However, despite the fact that these type of visualization techniques significantly reduce the amount of occlusions when the objects overlap, they appear less frequently in the literature when object alignment in AR/VR environments is required.

Despite the fact that semitransparent and wireframe techniques are frequently used in AR applications for object alignment, to the best of our knowledge, no previous work exists that evaluates which visualization technique is most suitable to support users to precisely align objects. Derived from this, we present a comparative evaluation of four visualization techniques used to render virtual objects during alignment tasks. The selection of these techniques is based on the amount of occlusion generated when the objects overlap. In this

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work, we propose the use of two visualizations that provide useful visual cues to the users with a very low amount of occlusions – *Fresnel-Derivative* and *Silhouette*–, and we compare them against two of the most used visualization techniques in previous works –*Semitransparent* and *Wireframe*– (Fig. 1).

We expect that users can perform significantly better in terms of alignment and time to completion when using visualization techniques that reduce the amount of occlusions when the objects overlap. Furthermore, we expect that the selection of visualization technique can play a determinant role in how the users perform and interact, specifically when the objects of interest are not designed to be assembled, or to perfectly match their counterparts. Examples of these applications are positioning devices with multiple mobile parts –e.g. medical X-Ray machines– or in applications for interior design.

To evaluate how the selected techniques affect users performance in alignment tasks, we developed an experimental setup in VR that allows us to have full control over the environment, ensuring that the objects used during alignment match the geometrical properties of their virtual representations. Our experimental design takes the canonical manipulation tasks presented by Bowman et al. [2] and the body-relative interaction modes for virtual object docking proposed by Mine et al. [15] into consideration.

In Section 2, we present a review of related work on applications that use visualization techniques in AR for tasks that require object alignment. The design considerations for the visualization techniques used in our experimental setup are described in Section 3. Technical details considered for the experiment design, user study, and data collection are shown in Section 4. In Section 5 we present a statistical analysis of quantitative measurements for alignment tasks in VR environments with 6 DoF –distance, rotation and time to completion–, as well as, qualitative scores –usability and mental effort–, derived from the implementation of a user study. Results suggesting that visualization techniques with low amounts of occlusion are better accepted and can help users to improve their performance during alignment tasks are discussed in Section 6. Lastly, we present our conclusions in Section 7.

2 RELATED WORK

One of the first attempts in demonstrating the usefulness of AR for assembling, manufacturing, and maintenance was introduced by Caudell and Mizell [5]. In this work, four applications were developed to improve human-involved operations in aircraft manufacturing using a see-through head-mounted display (HMD); one of such applications was designed to overlay 3D models of parts to be removed from the aircraft rendered as wireframes. Similarly, Feiner et al. [7], developed an AR-based system for printer maintenance and used a wireframe visualization technique to represent virtual parts to users using HMDs. Reiners [19] used AR for doorlock assembly in automotive scenarios in which CAD models were rendered as solid objects due to occlusion information was required. Zauner et al. [24], used solid and semitransparent virtual objects to provide step-by-step instructions to users in furniture assembly tasks. While these systems were intended to demonstrate the viability of using AR as a proof of concept in real life applications, no comparison of visualization techniques was conducted.

More recently, Oda et al. [16] presented a system to guide users to perform alignment tasks in 6 DoF. In this work, virtual 3D models were rendered as solid textured objects and presented to the users together with metaobjects to indicate where the real parts involved in the task had to be placed. To avoid occlusion problems, the virtual replica was faded out when the physical counterpart started to overlap, while the metaobjects remained visible to users. Task performance was evaluated in terms of time and not in terms of accuracy, being a task considered as completed when the differences in position and orientation were smaller than a threshold.

Similarly, Westerfield et al. [23], developed a framework for intel-

ligent AR training systems for computer motherboard assembly. In this work, 3D models of the parts to be installed in the motherboard were generated and shown as single-color semitransparent objects to guide the user through the process. Quantitative scores such as number of errors and total completion time, as well as qualitative results such as Likert-scale based questionnaires, were collected.

Other approaches used simpler models such as building blocks to demonstrate the usability of AR in assembly tasks. Tang et al. [21], evaluated the effectiveness of AR for object assembly when compared against common instructional media. The virtual models used in this work were rendered as semitransparent objects to indicate users where to place the blocks and time to completion, number of errors, and perceived mental workload were assessed.

Khuong et al. [12] compared two strategies to guide users during assembly tasks with Duplo blocks: the first one rendered virtual blocks represented as wireframes superimposed over the real structure, while the second one rendered a virtual replica of the building blocks as solid objects next to the real structure to avoid occlusions. Robertson et al. [20] combined semitransparent and wireframe techniques for rendering virtual objects to evaluate the effects of graphical context for assembly tasks using Lego blocks.

Alternative visualization techniques, such as the one proposed by Cole et al. [6] suggest that people is able to interpret shapes accurately when looking at line drawings. Kalkofen et al. [11] emphasized the importance of preserving context information to communicate spatial relationships in AR applications. To achieve this, the object edges were used as key features to provide depth cues to the augmentations while the remaining object surface was rendered fully transparent. Although this visualization technique was not used for object alignment, results demonstrate that Focus and Context techniques [13], enable users to draw the attention to objects in the focus while still perceiving contextual information.

Several studies conducted in VR environments, such as the one presented by Mine et al. [15] evaluated how body-relative interactions affect user performance during object docking tasks. Their study showed that direct manipulation techniques give users a greater sense of the position and orientation of the objects. Users performed best when the object to be manipulated is co-located with their hand. Jacob et al. [10] investigated the influence of input devices for interactive tasks, confirming the importance of matching the perceptual structure of the task to the control structure of the input device. Bowman et al. [2] highlighted the importance of the device shape for manual control tasks and presented a classification for canonical manipulation tasks in 3D, designated as basic manipulation tasks: *Selection*, *Positioning* and *Rotation*.

Despite the fact that many studies explored and tried to prove the effectiveness of AR/VR technologies for object alignment, there seems to be no reason in terms of accuracy and user interaction that justifies the author's selection of the visualization techniques. To the best of our knowledge, no previous work has tried to explore and evaluate how these visualization techniques affect the precision of object alignment in 6 DoF, how it influences user interaction, and the acceptance of the system.

3 METHODS

In order to evaluate how the selection of a visualization technique affects users' performance in tasks that require precise alignment in AR/VR, we developed a system in which users were asked to align textured objects¹ –hereinafter referred to as *dynamic objects* (Fig. 2)– with a position and orientation indicated by a virtual replica –hereinafter referred to as *static objects*– rendered using one of four visualization techniques described in Sect. 3.1.

We decided to design a fully controlled VR environment in which the geometric properties of the objects to align will be identical, so

¹ The Skull, Camera and Mug models are publicly available at: <https://free3d.com/>, the Balloon can be retrieved from: <https://docs.unity3d.com/Manual/SL-VertexFragmentShaderExamples.html>



Figure 2: Dynamic objects used to evaluate users performance during our study. From left to right: *Skull*, *Camera*, *Mug*, and *Balloon*.

we could evaluate users' performance while allowing us to acquire reliable measurements of position, orientation, and time. An example of the visual results obtained when the dynamic and static objects overlap is shown in Fig. 3.

3.1 Selection of Visualization Techniques

Since we expect that occlusions generated when the static and dynamic objects overlap influence users performance in alignment tasks, we decided to select and compare four visualization techniques with different transparency densities. In order to compare them, we rendered all the static objects using a single color. Furthermore, to evaluate the potential use of these techniques in AR applications, we simulated the conditions in which dynamic objects are not tracked by rendering the static objects always in front of the dynamic ones, additionally, the dynamic objects were not affected by shadows generated by the static objects. More details and design consideration for these techniques are described as follows:

3.1.1 Semitransparent

In this technique we rendered an object using a single color with an alpha value lower than 1.0. Modifying the alpha value allows users to observe the dynamic object through the static, even when occlusion is presented. Despite the fact, results from [4] suggest that alpha values between 0.6 and 0.8 are preferred by users, for our application these levels turned to be slightly high, leading to confusion during alignment, as a result of this, we selected an alpha value of 0.5. This visualization technique is based on the shader available at [18].

3.1.2 Wireframe

With this technique, we presented the static object as a single color mesh by drawing the edges of the model's faces using solid lines. As a result of this, the amount of occlusions generated by this technique strongly depends on the density of vertices the model contains, however, when the rate of occlusion is high, a reduction in the number of model vertices represents an alternative to decrease the mesh complexity without compromising the rendering quality and precision required for alignment. A similar approach was used in [19], with the exception that virtual models were rendered as solid objects and used for instructional purposes. This technique is based on the shader available at [14].

3.1.3 Fresnel-Derivative

We designed this technique by the addition of two core components. The first one allow users to observe the object edges and surfaces with high angles of incidence. The inverse additive of the dot product between the normal surfaces of the object and the users view direction –angle of incidence– is used to modify the alpha value in the models surface. The second element allow users to observe the model surfaces with high curvature by comparing the gradient of the object's surface and a threshold d_t . A similar approach to this technique was presented by Bichlmeier et al. [1] to improve perception of medical imaging data using AR. A mathematical representation of this model is described by equation 1.

$$\alpha = I_v * (d_f + d_c) \quad (1)$$

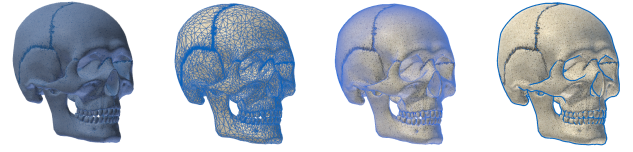


Figure 3: Example of a dynamic object after overlapping with each one of its static objects. From left to right: *Semitransparent*, *Wireframe*, *Fresnel-Derivative*, and *Silhouette*.

where α is the value of the objects alpha channel, I_v is a multiplicative intensity factor, while d_f , and d_c are the Fresnel and derivative components respectively. These components are computed using the equations 2 and 3.

$$d_f = (1.0 - \delta)^{f_p} \quad (2)$$

where δ is angle of incidence and f_p is the Fresnel factor power.

$$d_c = \begin{cases} 0 & \text{if } \left(\frac{d}{dx} SN + \frac{d}{dy} SN \right) \leq d_t \\ \frac{d}{dx} SN + \frac{d}{dy} SN & \text{if } \left(\frac{d}{dx} SN + \frac{d}{dy} SN \right) > d_t \end{cases} \quad (3)$$

where SN is object surface normal and d_t is a derivative threshold.

The parameters used during our user study were: $f_p = 17.5$ and $d_t = 0.215$.

3.1.4 Silhouette

For this visualization technique, we used two rendering phases. In the first phase, the objects surfaces were extended in the direction of their normals by a scale factor s_f and rendered using a solid color, producing an upscaled version of the model. In our study, the value of s_f was set to an additional 0.015 times the model size. A second phase rendered all the pixels of the model without growth as a fully transparent object. Resulting elements were rendered as solid lines with an alpha value of 1.0, while the rest of the object's surface was rendered with an alpha value of 0.0.

The combination of these phases produced the visualization of the object edges and some characteristic features that provided additional visual cues about the objects geometry. This visualization technique used as a base the shader available at [22].

3.2 Selection of Models

To highlight the properties of the visualization techniques described in Sect. 3.1, we selected four shapes with varying characteristics in terms of surface curvature and face density (Fig. 2). The *Mug model* has a low density of faces and a smoothly curved, homogeneous surface. Its handle is the main cue for orientation alignment. The *Camera model* has a curved and planar surface as well as a higher density of faces than the *Mug*. The *Skull model* consists of a combination of smooth and curved surfaces, sharp edges and has a higher face density than the previous models. Lastly, the extremities of the *Balloon model* are useful for precise alignment of the object's orientation, while its face density and curvature is similar to the *Skull model*.

As shown in Fig. 3, the density of the faces affects the occlusion rate for the *Wireframe technique*. High curvature regions of the models generate higher levels of occlusion when using the *Fresnel-Derivative technique*, while sharp edges are more visible when using the *Silhouette technique*.

4 USER STUDY

To evaluate the performance of the visualization techniques, we developed a VR application using the HTC Vive system² and Unity

²<https://www.vive.com/>

engine³. A pilot study served to validate and refine the design considerations of our system and a subsequent user study was conducted in order to collect the data used to compare the visualization techniques. This experiment was conducted in a private office space where environmental distractions were minimal.

Our experimental setup is composed of three different stages consisting of a tutorial and two test scenarios considering time constrained and unconstrained tasks from which data regarding position, orientation, and time were collected. After finishing the experiment, users were asked to answer a usability survey. The total duration of the experiment was between 40 and 60 minutes.

The technical aspects involved in the application design and the user study are described in this Section.

4.1 Participants

A total of thirty-two users (11 female and 21 male), all of them unpaid volunteers aged from 22 to 36 with a mean age of 27.03 ± 3.14 years gave their consent to take part in the experimental protocol. Over the 32 participants, 6 reported using AR/VR systems at least once a month and 19 never used the HTC Vive system (or similar) before. None of the participants had previous experience using our system.

4.2 Experimental Setup

As a prerequisite, participants were requested to perform the standard Ishihara test [9] for detection of color vision deficiency using a personal computer. After finishing the Ishihara test, the HTC Vive headset was provided to the users and a virtual screen showing an introductory message was shown. Participants were asked to adjust the HMD lenses to their interpupillary distance until they were able to read the text clearly. All participants wearing glasses to correct vision problems were allowed to use them during the experiment.

4.2.1 Interaction Modes

Users were able to translate and rotate the dynamic objects with a HTC Vive controller using two interaction modes: *Normal mode* allowed users to manipulate the dynamic object using a direct attachment to the interaction handle. *Precise mode* was only enabled when the Euclidean distance between the static and dynamic object was below 10 centimeters and mapped the relative translation and orientation changes of the interaction handle with a ratio of ten-to-one onto the dynamic object. Both modes were controlled by separate buttons. A third button was used to confirm the alignment, which had to be pressed for two seconds in order to avoid false positives.

4.2.2 Tutorial

In order to provide participants with the ability to manipulate the dynamic objects, they were asked to perform a tutorial session in which they interacted with dynamic and static objects. Neither the object nor the visualization technique used in the tutorial were used for the experiment. The three interaction modes described in Sect. 4.2.1 were introduced during the tutorial. Participants were allowed to interact freely with the tutorial until they felt comfortable manipulating the objects in the scene.

4.2.3 Object Alignment with No Time Constraints (NTC)

For the NTC stage, user were asked to align as precisely as possible a dynamic object in the position and orientation defined by the static object. A total of 16 different scenarios for alignment were presented to the users resulting from the combinations of objects (Fig. 2) and visualization techniques (Fig. 1). Only one pair of dynamic and static objects was shown at a time. Participants were allowed to take as much time as needed to achieve proper alignment and after user confirmation, a new pair of objects were shown. After completion of the NTC scenario, users had the option to take a short break.

4.2.4 Object Alignment with Time Constraints (TC)

This stage of our study considered a scenario in which object alignment had to be achieved within a certain amount of time. The objects and techniques were the same as those used in the NTC stage, however, the order of appearance for the objects was different.

Users were instructed to keep aligning the objects as precisely as possible and they were notified about the time constraints. Participants were allowed to evaluate the scene before moving the objects, however, any change in the pose of the dynamic object triggered the countdown of a timer that was always visible. Once this timer ran out of time, a new pair of objects were shown in the scene.

4.2.5 System Usability Score (SUS) and Mental Effort (ME)

After completion of the NTC and TC scenarios, participants were asked to evaluate the visualization techniques using the system usability score (SUS) proposed by [3], as well as to report the mental effort perceived (ME) using the 9-point scale introduced by Paas [17].

4.3 Parameters Selection

As a result of a pilot study performed before the user study, we decided to locate all objects at a height of 1.35 meters. The dynamic objects were placed in the center of the scene, while the static objects appeared pseudo-randomly on a circumference of a circle with radius equal to 0.75 meters with center in the dynamic object. The orientation of the static objects was also defined pseudo-randomly. We deliberately chose an empty virtual scene with a homogeneous, gray colored background to avoid unwanted interference with background clutter or distraction.

To mitigate learning effects, a 16×16 Latin square matrix was used to select the combination of visualization technique and object type presented to participants. The time limit value for the TC scenario was set to 30 seconds, which correspond to the time required for most of the participants in the pilot study to achieve an acceptable level of alignment under NTC conditions.

4.4 Experimental Variables

For the experimental setup, two independent variables –each with four different levels– corresponding to the model and visualization technique were involved in both of our scenarios –NTC and TC–. For the NTC scenario, three dependent variables were computed after confirmation. Position errors were calculated as the Euclidean distance between the gravity centers of the dynamic and static objects. Orientation errors in degrees were computed using an axis–angle transformation that rotates the dynamic object to match the orientation of the static object. The time to completion was considered as the time elapsed since the first user interaction with the dynamic object until confirmation. For the TC scenario, only the variables for position and orientation were considered.

4.5 Hypotheses

Taking into consideration the occlusion problems described in Section 1 we formulated the following hypotheses:

- H1.** Users perform better in terms of position, orientation and time to completion under NTC conditions when visualization techniques with low rates of occlusion are used compared to those that present high levels of occlusion.
- H2.** Users perform better in terms of position and orientation under TC conditions when visualization techniques with low rates of occlusion are used compared to those that present high levels of occlusion.
- H3.** Users preference in terms of usability and mental effort is higher when visualization techniques with low rates of occlusion are used compared to those that present high levels of occlusion.

³ <https://unity3d.com/>

Table 1: Mean and standard deviation values registered by users after grouping data results by visualization technique.

Condition	Distance (in meters)				Rotation (in degrees)				Time (in seconds)		Usability			
	NTC		TC		NTC		TC		NTC		SUS		ME	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Fresnel-Derivative	0.0026	0.0017	0.0042	0.0042	2.065	1.764	3.487	3.058	84.98	58.60	67.34	22.58	4.69	2.22
Semitransparent	0.0025	0.0017	0.0038	0.0030	1.917	1.362	2.983	2.504	84.45	47.34	59.45	17.61	5.97	2.02
Silhouette	0.0022	0.0015	0.0041	0.0040	1.790	1.299	2.610	1.907	78.12	48.15	75.78	18.53	4.28	1.89
Wireframe	0.0030	0.0030	0.0044	0.0040	2.165	1.467	3.222	2.689	92.75	55.52	63.44	25.84	5.00	2.59

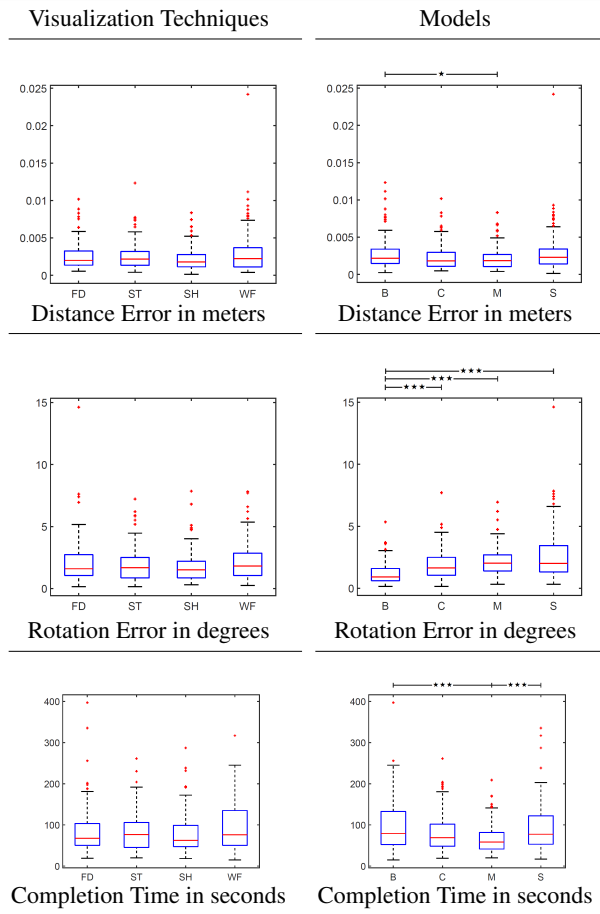


Figure 4: Scores obtained by users under NTC conditions. (Left) Results grouped by visualization techniques: *FD-Fresnel-Derivative*, *ST-Semitransparent*, *SH-Silhouette*, *WF-Wireframe*. (Right) Results grouped by model: *B-Balloon*, *C-Camera*, *M-Mug*, *S-Skull*. (The red line indicates the median and the bottom and top edges of the blue box indicate the 25th and 75th percentiles).

5 RESULTS

In this Section, we provide a statistical analysis of the results collected in our study for the NTC and TC scenarios, as well as for SUS and ME scores. Since the collected data is not normally distributed, we used a Kruskal-Wallis test with $\alpha = 0.05$ to compare the results obtained in terms of position and orientation for the NTC and TC scenarios, as well as for time to completion in the NTC scenario. Similarly, this test was used to evaluate the scores obtained by SUS and ME. Posterior Bonferroni tests were used to reveal significant differences between the variables based on the results obtained from the Kruskal-Wallis tests. Significant differences between variables are indicated as $\star$ ($p < 0.05$), $\star\star$ ($p < 0.01$), and $\star\star\star$ ($p < 0.001$) in Fig. 4 (NTC), Fig. 5 (TC), and Fig. 6 (SUS and ME).

Overall, the Silhouette technique obtained the best scores for

position, orientation and time to completion under NTC condition, for rotation under TC, and for usability and mental effort. The Semi-transparent technique scored best in terms of translation for the TC scenario. Additional analysis in terms of model behavior revealed the highest scores for position when the Mug was used, while the Balloon presents the best scores for orientation. During data analysis, one outlier with an orientation error of 169.52 degrees corresponding to the Skull model rendered using the Wireframe visualization technique was replaced with the mean value of the sample.

5.1 NTC Statistical Analysis

Analysis of the data collected in terms of visualization techniques, report the lowest errors in terms of mean and standard deviation for the Silhouette visualization technique and the highest for the Wireframe, however, Kruskal-Wallis tests did not reveal significant interaction between these techniques. Contrastingly, statistically significant differences between models for distance scores ($\chi^2(3) = 13.01, p < 0.01$) were found. Participants were able to position the Mug significantly better than the Balloon ($p < 0.05$).

In terms of rotation, the mean scores achieved by users reveal the lowest errors for the Silhouette visualization technique and the highest for the Wireframe, however, Kruskal-Wallis results failed to reveal significant differences for the visualization techniques. Similarly to the position score, statistically significant differences between models were found for rotation ($\chi^2(3) = 79.21, p < 0.001$). Participants achieved significantly better scores for aligning the orientation of the Balloon than for the Camera ($p < 0.001$), the Mug ($p < 0.001$), or the Skull ($p < 0.001$).

Lastly, and similarly to the results obtained for position and orientation, the Silhouette technique obtained the best results in terms of time to completion, while the Wireframe performed the worst. However, Kruskal-Wallis results revealed no significant difference between visualization techniques for the time scores. In terms of models, Kruskal-Wallis results revealed statistically significant difference for the time scores ($\chi^2(3) = 23.74, p < 0.001$). Users completed significantly faster the alignment task when aligning the Mug compared to the Balloon ($p < 0.001$) and the Skull ($p < 0.001$).

5.2 TC Statistical Analysis

Similarly to the NTC scenarios, Kruskal-Wallis and Bonferroni post-hoc tests were used to reveal statistical significance between the models and visualization techniques. In terms of distance, no significant difference between visualization techniques for the TC scores was revealed. However, significant differences between models ($\chi^2(3) = 41.3, p < 0.001$) were found. Participants were able to align better the Mug compared to the Balloon ($p < 0.001$) and the Skull ($p < 0.01$), as well as the Camera compared to the Balloon ($p < 0.01$) and the Skull ($p < 0.05$).

In terms of rotation, participants were able to achieve better scores using the Silhouette, however, Kruskal-Wallis results revealed no significant difference between visualization techniques. Additionally, significant difference between models for the rotation scores ($\chi^2(3) = 61.6, p < 0.001$) were found. Users performed better in terms of orientation aligning the Balloon compared to the Camera ($p < 0.001$), the Mug ($p < 0.001$), and the Skull ($p < 0.001$); and using the Camera compared to the Skull ($p < 0.01$).

Table 2: Mean and standard deviation values obtained by users after grouping data results by model.

Condition	Distance (in meters)				Rotation (in degrees)				Time (in seconds)	
	NTC		TC		NTC		TC		NTC	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Balloon	0.0028	0.0021	0.0049	0.0040	1.199	0.859	1.976	1.660	96.22	58.81
Camera	0.0023	0.0017	0.0039	0.0043	1.919	1.205	2.913	2.375	83.64	50.19
Mug	0.0022	0.0015	0.0029	0.0025	2.196	1.177	3.238	2.241	67.80	38.15
Skull	0.0030	0.0028	0.0048	0.0040	2.624	2.058	4.181	3.318	92.76	57.19

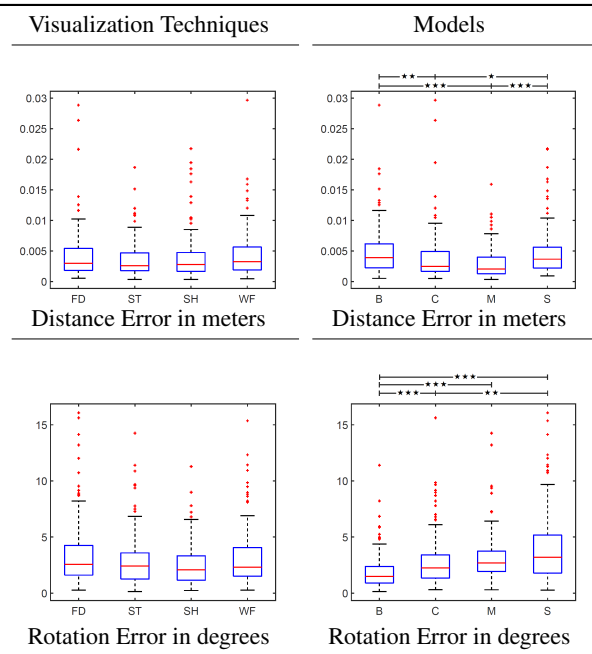


Figure 5: Scores obtained by users under TC conditions. (Left) Results grouped by visualization techniques: *FD-Fresnel-Derivative*, *ST-Semitransparent*, *SH-Silhouette*, *WF-Wireframe*. (Right) Results grouped by model: *B-Balloon*, *C-Camera*, *M-Mug*, *S-Skull*. (The red line indicates the median and the bottom and top edges of the blue box indicate the 25th and 75th percentiles).

5.3 SUS and ME Statistical Analysis

In terms of perceived usability and mental effort, the Silhouette and Fresnel-Derivative techniques obtained the highest scores, followed by the Wireframe and Semitransparent. SUS scores between 51 and 68 are considered average results, scores between 68 and 80.3 are above average, and results higher than 80.3 are considered in the top 10% scores. Mean and standard deviation results for the SUS tests are summarized in Table 1 and show that the Silhouette technique was the only visualization technique that scored above 68 but under 80.3. Kruskal-Wallis results revealed significant differences between visualization techniques for the SUS score ($\chi^2(3) = 10.26, p < .05$). A Bonferroni post-hoc test revealed that usability scores were significantly better for the Silhouette visualization technique compared to the Semitransparent ($p < 0.05$).

Mental effort results are similar to the SUS scores, being Silhouette and Fresnel-Derivative the visualization techniques that presented the lowest mental effort, while Wireframe and Semitransparent scored the highest values. Our Kruskal-Wallis results revealed significant difference between visualization techniques for mental effort ($\chi^2(3) = 9.88, p < .05$). A post-hoc test revealed that the mental effort using the Silhouette technique was significantly lower when compared to Semitransparent visualization technique ($p < 0.05$). Summarized results of these scores are shown in Fig. 6 and Table 1.

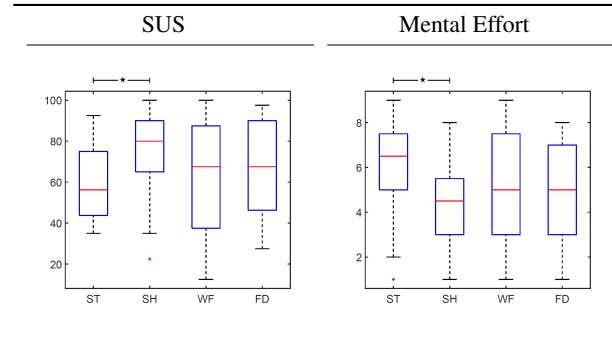


Figure 6: Scores for SUS and Mental Effort reported by users. *ST-Semitransparent*, *SH-Silhouette*, *WF-Wireframe*, *FD-Fresnel-Derivative*. (The red line indicates the median and the bottom and top edges of the blue box indicate the 25th and 75th percentiles).

6 DISCUSSION

In terms of object geometry, two main tendencies can be observed. Rotation results show evidence that the Balloon model obtained the best scores for orientation. This can be related to the length of the object extremities that provide visual cues of misalignment when the rotation of the dynamic and static objects are not similar. This can also explain the reason why objects with relatively high curved surfaces and without long extremities, such as the Mug and the Skull, present the worst scores in terms of rotation. Contrastingly, in terms of position, the Mug appears to be the object with better results, while the Balloon shows to be the worst. This effect may lead to suppose there is a relation in how the geometrical properties of the objects lead users to concentrate their attention in aligning one of this two parameters—position or rotation—underestimating the other. However, a deeper analysis in this regard should be conducted.

Another interesting effect observed in terms object geometry is the fact that, despite the increment in error values for the TC scenario with respect to the NTC, similar statistical significance was observed for both position and orientation. This pattern may suggest that users performance over time is not related to the geometry of the object.

With regard to the results obtained by comparing the visualization techniques, the Silhouette technique performed better in 4 out of 5 accuracy scores: position (NTC), orientation (NTC and TC), and time, and only the Semitransparent technique performed better for distance (TC). However, no statistical significance was observed after Kruskal-Wallis tests. The lack of differences in the results obtained under NTC conditions may be explained as a result of the amount of time available for users to align the objects. Nevertheless, the behavior was similar to the one observed in the TC scenario.

Considering the results of our study under the NTC condition, **H1** is partially supported by the results obtained with the Silhouette technique in terms of position, orientation, and time, while **H2** is not supported considering the results obtained by the Semitransparent technique. The scores of usability and mental effort achieved by the Silhouette technique were significantly better when compared to the Semitransparent technique, and therefore supporting **H3**.

7 CONCLUSIONS & FUTURE WORK

In this paper, we presented a user study to compare the performance of visualization techniques for precise object alignment in tasks that require 6 DoF in VR environments. The selection of techniques relies on the amount of occlusions observed when the objects to be aligned overlap. Our results suggest that using visualization techniques with a low amount of occlusion may lead to better performance for position, orientation, and completion time. However, no statistically significant difference was obtained to fully support the aforementioned hypotheses. Additionally these results suggest that users prefer the techniques that present the lowest amount of occlusion in terms of usability and mental effort. Results obtained for the selected geometries suggest a relation between the geometrical properties of the object to be aligned and the performance achieved by users.

We deliberately reduced the complexity of our virtual environment using a homogeneous background to ensure that the collected data genuinely reflects the effects of static visualization techniques on the users performance during alignment tasks. This enables a precise evaluation of the effects of object complexity and visualization technique while reducing the influence of the surrounding environment. The performance evaluation in this paper represents a necessary first step to better understand the benefits of static visualization techniques for object alignment, which work without precise tracking of the dynamic object.

In future work, we would like to explore how additional factors such as scenarios with cluttered or dynamic backgrounds, as well as limitations of display technologies for AR systems might affect the user's performance and preference. Furthermore, as a result of the observations made during the participation of users in our study, we plan to investigate if the selection of visualization technique affects the user's behaviour during task performance, helping to reduce the motion parallax or improving the depth perceived by users through analyzing their viewpoint trajectory. Additional performance improvements could be found by selecting the appropriate visualization technique during a task to optimally match spatial relations, context, and the user's experience.

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